Dr. Chandan Patra

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Education

- **Post Doctoral Research Associate**, S. N. Bose National Centre for Basic Sciences, Joined on 21st February, 2024.
- **Ph.D. thesis on "Unconventional Superconducting Properties of Layered Chalcogenides"**, submitted to the Indian Institute of Science Education and Research Bhopal (IISER Bhopal), completed on 28th December 2023.
- **M.S. in Physics**, Indian Institution of Science Education and Research Bhopal (IISER Bhopal), 2016-2019, achieved 8.61 out of 10 on a CPI basis.
- **B.Sc. in Physics**, Vidyasagar University (Midnapore College), 2013-2016, achieved 70.60%.
- **Higher Secondary(10+2)**, West Bengal Council of Higher Secondary Education, 2011-2013, achieved 82.00% (top 1% of the board).
- **Secondary (10)**, West Bengal Council of Secondary Education, 2009-2011, achieved 81.70%.

Qualified National Examinations

- **Joint Entrance Screening Test (JEST) (2016).**
- **Joint Admission Test for M.Sc. (JAM) (2016).**
- **Graduate Aptitude Test in Engineering (2020).**

Teaching Experience

- **PHY-103 : General Physics Laboratory-I, 2019-2020.**
- **PHY-101 : Mechanics, 2020-2021.**
- **PHY-406 : Nuclear Physics Laboratory, 2020-2021.**
- **PHY-402 : Atomic and Molecular Physics, 2021-2022.**

Fellowships

- **Senior Research Fellowship, IISER Bhopal, India (2021-2023).**
- **Junior Research Fellowship, IISER Bhopal, India (2019-2021).**
- **Institute Fellowship (MS Course), IISER Bhopal, India (2016-2019).**
- **Inspire Scholarship, India (2013-2016).**

Research Interest

- ★ **Low dimensional superconducting materials.**
- ★ **Topological superconductor.**
- ★ **Strongly correlated electron system.**
- ★ **Magnetism and magnetotransport in 2D van der Waals (vdW) materials.**

Current Research

My research primarily focused on investigating the superconducting and normal state (planar Hall) properties of **single crystal** of transition metal dichalcogenides through the introduction of controlled disorder. Recent observations have indicated that this controlled disorder can enhance superconductivity by suppressing charge density waves and enhancing electron-phonon coupling. This approach holds the potential to offer new insights into the intricate relationship between charge density waves, superconductivity, and disorder. In the case of the $2H$-TaSeS disorder-controlled system, we have observed anisotropic multigap behavior. Additionally, within the NbSeTe system, the observed planar Hall effect indicates the presence of topological superconductivity. Furthermore, another intriguing system RhSeTe with coherent disorder reveals a type-II Dirac point located above 82 meV from the Fermi surface, as determined from density functional theory (DFT) calculations. In addition, I have gained work experience in the field of two-dimensional van der Waals ferromagnetism. Co doping in Fe_3GeTe_2 induces significant changes in the Hall effect and results in the formation of a skyrmion lattice structure, as observed through magneto-optical Kerr effect (MOKE) imaging.

Publications

12. Two-dimensional multigap superconductivity in bulk $2H$-TaSeS; **C. Patra**, T. Agarwal, Rajeshwari R. Chaudhari, and R. P. Singh, Phys. Rev. B 106, 134515 (2022).

11. Planar Hall effect and quasi-2D anisotropic superconductivity in topological candidate $1T$-NbSeTe; **C. Patra**, T. Agarwal, S. Srivastava, R. R. Chowdhury, M. P. Saravanan, R. P. Singh, Adv. Quantum Technol. 2300448 (2023).

10. Superconducting Properties of Topological Semimetal $1T$-RhSeTe; **C. Patra**, T. Agarwal, Arushi, P. Manna, N. Bhatt, R. S. Singh, R. P. Singh, arXiv:2311.01019v1 (2023), under review in Physical Review B.

9. Skyrmion lattice structures and modification of emergent magnetic field with Co-doping in uniaxial van der Waals Fe_3GeTe_2; Rajeswari Roy Chowdhury, Samik DuttaGupta, **Chandan Patra**, Oleg A. Tretiakov, Sudarshan Sharma, Shunsuke Fukami, Hideo Ohno, and Ravi Prakash Singh, Scientific reports 11, 1 (2021).

8. Modification of unconventional Hall effect with doping at the nonmagnetic site in a 2D van der Waals ferromagnet; Rajeswari Roy Chowdhury, **Chandan Patra**, Samik DuttaGupta, Sayooj Satheesh, Shovan Dan, Shunsuke Fukami, and Ravi Prakash Singh, Phys. Rev. Materials 6, 014002 (2022).

7. Anisotropic magnetotransport in the layered antiferromagnet $TaFe_{1.25}Te_3$; Rajeswari Roy Chowdhury, Samik DuttaGupta, **Chandan Patra**, Anshu Kataria, Shunsuke Fukami, and Ravi Prakash Singh, Phys. Rev. Materials 6, 084408 (2022).

6. Quasi-two-dimensional anisotropic superconductivity in Li-intercalated $2H$-TaS_2; T. Agarwal, **C. Patra**, A. Kataria, Rajeswari R. Chowdhury, and R. P. Singh, Phys. Rev. B 107, 174509 (2023).

5. Superconducting ground state of nonsymmorphic superconducting compound Zr_2Ir; Manasi Mandal, **Chandan Patra**, Anshu Kataria, D. Singh, P. K. Biswas, J. S. Lord, A. D. Hillier, and R. P. Singh, Phys. Rev. B 104, 054509 (2021).

4. Spin-polarized supercurrent through the van der Waals Kondo-lattice ferromagnet Fe_3GeTe_2; Deepti Rana, Aswini R, Basavaraja G, **Chandan Patra**, Sandeep Howlader, Rajeswari Roy Chowdhury, Mukul Kabir, Ravi P. Singh, and Goutam Sheet, Phys. Rev. B 106, 085120 (2022).

3. Superconductivity in doped Weyl semimetal $Mo_{0.9}Ir_{0.1}Te_2$ with broken inversion symmetry; Manasi Mandal, **Chandan Patra**, Anshu Kataria, Suvodeep Paul, Surajit Saha and R P Singh, Supercond. Sci. Technol. 35 025011 (2022).

2. Superconductivity in noncentrosymmetric NbReSi investigated by muon spin rotation and relaxation; Sajilesh K. P., K. Motla, P. K. Meena, A. Kataria, **C. Patra**, Somesh K., A. D. Hillier, and R. P. Singh, Phys. Rev. B 105, 094523 (2022).

1. Time-reversal symmetry breaking in frustrated superconductor Re_2Hf; Manasi Mandal, Anshu Kataria, **Chandan Patra**, D. Singh, P. K. Biswas, A. D. Hillier, Tanmoy Das, and R. P. Singh, Phys. Rev. B 105, 094513 (2022).

Research Experience

Hardware Skills	Software Skills
Physical Property Measurement System (PPMS), Quantum Design SQUID Magnetometer, X'pert Pro PANalytical diffractometer, Laue Diffractometer, High temperature box and tubular furnaces, Vacuum pumps and cryogenic Operating Systems, TEM, SEM, and EDAX, Glove Box, Single-arc furnace, Optical Floating zone, Tetra-arc furnace, Flux method, Chemical vapour transport method, Glassblowing and Vacuum sealing.	C, Fortran, Mathematica, Matlab, Origin and Igor, Full Prof and High Score Plus, Wimda and Mantid for μSR data analysis, Crystal Maker and Jana 2006, Windows and Linux, MS Office and LATEX.

Conferences and School

- ★ **UKRI Science and Technology Facilities Council (STFC) Neutron and Muon Source at the Rutherford Appleton Laboratory**, **Harwell Campus, Didcot, Oxfordshire**, UK. For Muon study on 2H-TaSeS single crystal on December 9-18th, 2023.

- ★ **KEK-IINAS School / The 5th Neutron and Muon School**, **J-PARC, Ibaraki**, Japan. (December 6-9th, 2021).

- ★ **Spectroscopies in Novel Superconductors (SNS), Indian Institute of Science Bangalore, India. (12-16th December 2022).**

- Pacific Rim International Conference on Superconducting Materials: Fundamentals and Applications - PRISM, ACCMS-Global Research Center SRMIST, Chennai India and New Industry Creation Hatchery Center, Tohoku University, Japan. September 22-23th (2022).
- In-house Symposium 2019 Department of Physics, IISER Bhopal, 24-25th January 2020.
- National Assembly of Researchers In Physics (NARIPHY), IISER Bhopal, 2022.

References

- **Dr. Makhanlal Nanda Goswami**
  Associate Professor,
  Department of Physics,
  Midnapore College (Autonomous), West Bengal, India
  Tel: +91-
  Email: makhanlal-nanda.goswami@midnaporecollege.ac.in

- **Dr. Tanusri Pal**
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Declaration

I, Chandan Patra, hereby declare that the facts given above are genuine to the best of my knowledge and belief.